

Completing the Components of Project Charter

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Context & Value

- What is a project charter?

Short document that officially starts a project.

Defines what the project aims to achieve and who is involved, helping align everyone before the project starts.

- Why it's critical in data engineering projects?

Alignment

Scope

Expectations

- How it sets the stage for successful execution of the capstone?

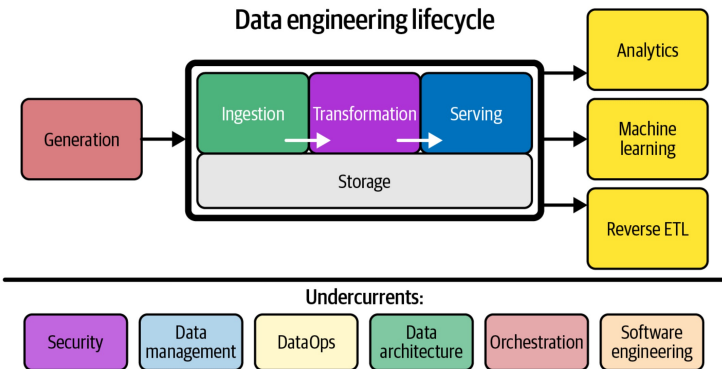
It creates a clear roadmap and ensures **everyone** **understands** what needs to be done, by whom, and why.

1. Project Overview & Objectives

- Brief description of the capstone.
- SMART goals
(e.g., process X TB of data, improve ETL runtime by 30%).

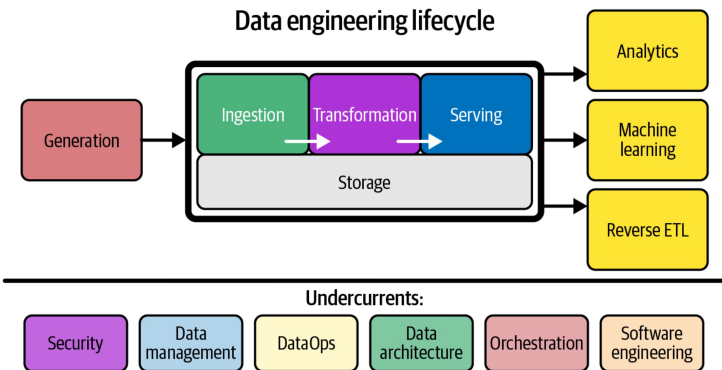
Letter	Meaning	Explanation
S	Specific	Clearly define what you want to achieve.
M	Measurable	Ensure progress can be tracked or quantified.
A	Achievable	The goal should be realistic and attainable.
R	Relevant	Should align with the project or overall objectives.
T	Time-bound	Set a deadline or time frame to complete the goal.

SMART Framework for Goal Setting



Generation: Data is created or collected from various sources such as applications, sensors, logs, or external systems.

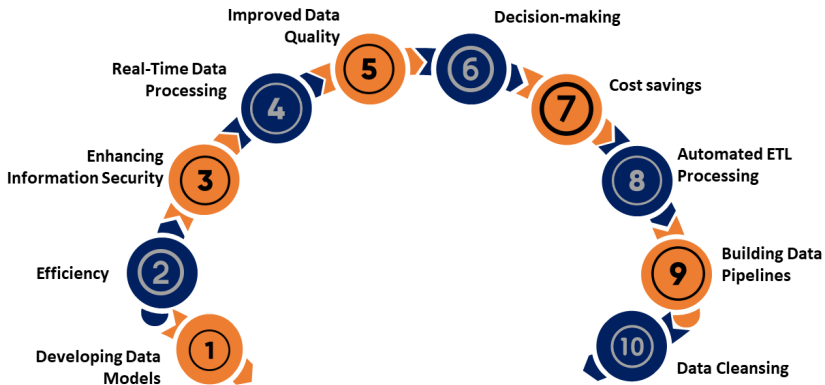
Ingestion: Data engineers build pipelines to bring data into the system—streaming or batch. This is where extraction and initial loading occur.



Transformation: Data is cleaned, standardized, enriched, and modeled. This stage ensures the data is usable and trustworthy.

Serving: Processed data is made available for consumers. This could be via databases, APIs, or data warehouses/lakes.

Advantages of Using Data Engineering Tools



2. Scope Definition

- In-scope vs out-of-scope components.
- Data sources, pipelines, and deliverables (dashboards, ML feature store, documentation).

THE DATA SCIENCE HIERARCHY OF NEEDS

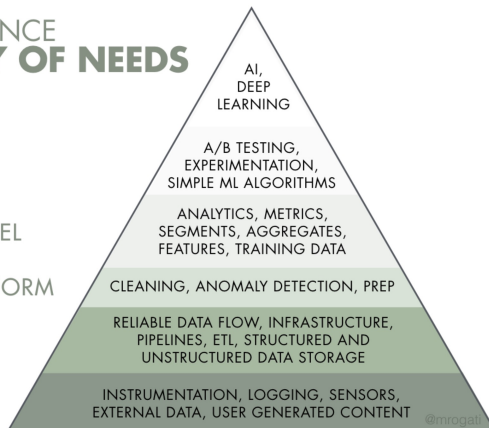
LEARN/OPTIMIZE

AGGREGATE/LABEL

EXPLORE/TRANSFORM

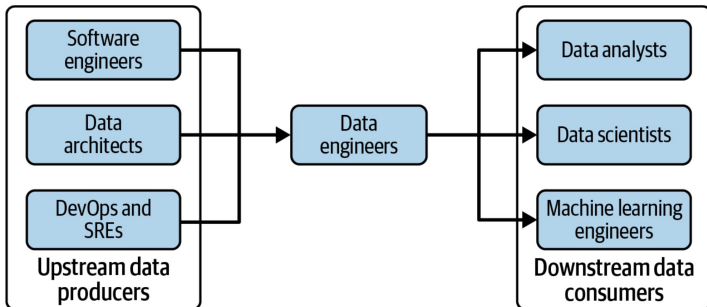
MOVE/STORE

COLLECT



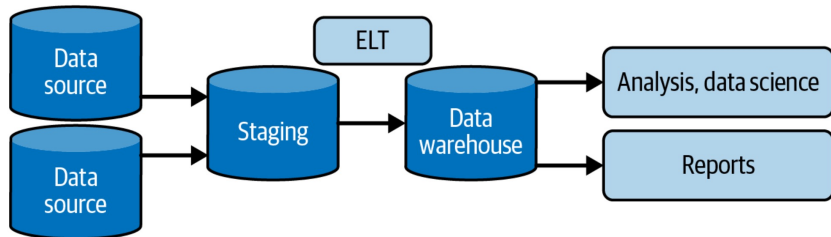
3. Stakeholders & Roles

- Project sponsor
- Technical lead
- Data engineer(s)
- Advisor / mentor
- End users



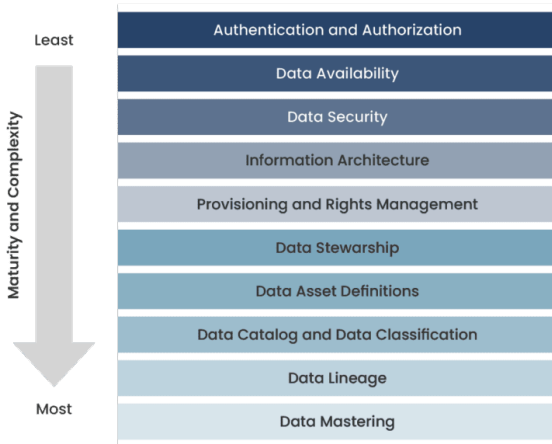
4. Project Deliverables

- ETL pipeline prototype
- Data model / schema
- Storage solution (e.g., warehouse, lake)
- Performance benchmarks
- Final presentation/document



6. Assumptions & Constraints

- Data availability
- Data standards
- Tools, time and resource limits



7. Risks & Mitigation

- Data quality issues → define cleanup strategy
- Performance limits → plan scaling options
- Dependency on external APIs → backup datasets

Steps to Conduct a Data Risk Assessment



Conclusion

"A clear charter leads to a clear path."

- Clarity and alignment
- Validate goals using data feasibility
- Ensure risks & constraints reflect real limitations

